Helicons

EM wave in electron gas subject to external magnetic field B_0

$$\varepsilon_{ij}(\omega) = \varepsilon_0 \delta_{ij} + \frac{i}{\omega}\sigma_{ij}(\omega)$$

We found
$$\sigma^{\perp}_{ij}(\omega) = \frac{e^2 n}{m}\begin{pmatrix} -i\omega + \frac{1}{\tau} & -\Omega_c \\ \Omega_c & -i\omega + \frac{1}{\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

$$\sigma_{zz}(\omega) = \frac{e^2 n}{m}\frac{1}{-i\omega + \frac{1}{\tau}} \quad ; \text{ also } \omega_p^2 = \frac{e^2 n}{\varepsilon_0 m}$$

$$\Rightarrow \frac{i\sigma_{zz}}{\omega} = \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})}$$

$$\frac{i\sigma^{\perp}}{\omega} = \varepsilon_0 \omega_p^2 \begin{pmatrix} 1 + \frac{i}{\omega\tau} & -\frac{i\Omega_c}{\omega} \\ \frac{i\Omega_c}{\omega} & 1 + \frac{i}{\omega\tau}\end{pmatrix}\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2}$$

We take the limit $\frac{1}{\tau} \ll \omega \ll \Omega_c \ll \omega_p$

Then
$$\frac{1}{\Omega_c^2 + (-i\omega + \frac{1}{\tau})^2} \approx \frac{1}{\Omega_c^2 - \omega^2} \approx \frac{1}{\Omega_c^2}$$

$$\frac{1}{\omega\tau} \ll 1.$$

$$\Rightarrow \varepsilon^{\perp}(\omega) \cong \cancel{\varepsilon_0 \delta_{ij}} + \varepsilon_0 \begin{pmatrix} \frac{\omega_p^2}{\Omega_c^2} & \frac{-i\omega_p^2}{\Omega_c \omega} \\ \frac{i\omega_p^2}{\Omega_c \omega} & \frac{\omega_p^2}{\Omega_c^2}\end{pmatrix}$$

and we can neglect $\varepsilon_0 \delta_{ij}$ in comparison with $\left(\frac{\omega_p}{\Omega_c}\right)^2 \gg 1$.

Also, $\varepsilon^{\parallel}(\omega) = \varepsilon_{zz}(\omega) = \varepsilon_0 \delta_{ij} + \varepsilon_0 \frac{i\omega_p^2}{\omega(-i\omega + \frac{1}{\tau})} \approx -\varepsilon_0 \frac{\omega_p^2}{\omega^2}$

Observe that $\frac{\omega_p^2}{\Omega_c \omega} = \left(\frac{\omega_p}{\Omega_c}\right)^2 \frac{\Omega_c}{\omega} \gg \left(\frac{\omega_p}{\Omega_c}\right)^2$

$$\Rightarrow \varepsilon_{xx} \ll \varepsilon_{xy} \ll \varepsilon_{zz}$$

$$\begin{cases} \vec{k}\times\vec{B} = -\mu_0\omega\vec{D} = -\dfrac{\omega}{\varepsilon_0 c^2}\vec{D} \\ \vec{k}\times\vec{E} = \omega\vec{B} \end{cases}$$

$$\Rightarrow \underbrace{\vec{k}\times(\vec{k}\times\vec{E})}_{\vec{k}(\vec{k}\cdot\vec{E})-\vec{E}k^2} = -\frac{\omega^2}{\varepsilon_0 c^2}\vec{D}$$

$$\Rightarrow \left(k^2\delta_{ij} - k_i k_j\right)E_j = \frac{\omega^2}{\varepsilon_0 c^2}\underbrace{\varepsilon_{ij}(\omega)E_j}_{D_i}$$

Let us consider wave propagating along $\vec{B}_0 \Rightarrow \vec{k} = (0,0,k)$
$\hookrightarrow B_0\hat{z}$.

$$\Rightarrow (k^2 - k_z^2)E_z = 0 \Rightarrow D_z = 0 \Rightarrow E_z = 0.$$

The wave is transverse $\vec{E} = (E_x, E_y, 0)$, $\vec{k}\cdot\vec{E} = 0$.

$$k^2 E_x = \frac{\omega^2}{\varepsilon_0 c^2}\left(\varepsilon_{xx}E_x + \varepsilon_{xy}E_y\right) =$$

$$= \frac{\omega^2}{\varepsilon_0 c^2}\varepsilon_0\left(\frac{\omega_p^2}{\Omega_c^2}E_x + \frac{-i\omega_p^2}{\Omega_c\omega}E_y\right)$$

$$\begin{cases} \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_x + \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_y = 0 \\ \left(k^2 - \dfrac{\omega^2\omega_p^2}{c^2\Omega_c^2}\right)E_y - \dfrac{i\omega\omega_p^2}{c^2\Omega_c}E_x = 0 \end{cases}$$

$$\begin{bmatrix} & \\ & \end{bmatrix}\begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0 \Rightarrow \det\begin{bmatrix} & \\ & \end{bmatrix} = 0$$

$$\Rightarrow \left(k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2}\right)^2 - \left(\frac{\omega \omega_p^2}{c^2 \Omega_c}\right)^2 = 0$$

$$k^2 - \frac{\omega^2 \omega_p^2}{c^2 \Omega_c^2} = \pm \frac{\omega \omega_p^2}{c^2 |\Omega_c|}$$

But $\frac{\omega_p^2}{c^2}\left(\frac{\omega}{\Omega_c}\right)^2 \ll \frac{\omega_p^2}{c^2}\frac{\omega}{\Omega_c}$

$\Rightarrow$ can neglect $\left(\frac{\omega \omega_p}{c \Omega_c}\right)^2$!

$$\Rightarrow \omega = c^2 k^2 \frac{\Omega_c}{\omega_p^2}$$ (only $\oplus$ sign is physical for $\Omega_c > 0$)

1) The obtained dispersion is quadratic, $\omega \sim k^2$. And the frequency is quite low since $\frac{\Omega_c}{\omega_p} \ll 1$.

2) $$\begin{pmatrix} k^2 & \frac{i\omega \omega_p^2}{c^2 \Omega_c} \\ -\frac{i\omega \omega_p^2}{c^2 \Omega_c} & k^2 \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} = 0$$

$$\Rightarrow E_x = -\frac{i\omega \omega_p^2}{c^2 k^2 \Omega_c} E_y = -i E_y \quad \text{for } \Omega_c > 0$$

(and $E_x = i E_y$ for $\Omega_c < 0$)

$\Rightarrow$ the propagating EM wave is circularly polarized!

Right-handed polarization for $\Omega_c > 0$ (electrons)

$\rightarrow$ helicon (in plasma physics $\rightarrow$ whistling atmospherics)